

PREFACE FOR FACULTY

The Decimation-in-Frequency (DIF) FFT algorithm (also known as the Sande-Tukey algorithm) provides an alternative decomposition of the DFT by dividing the frequency spectrum $X[k]$ into even and odd frequency bins.

Pedagogical Strategy:

1. Derive DIF decomposition by splitting the time-domain input summation into first and second halves:
 $n \in [0, N/2 - 1]$ and $n \in [N/2, N - 1]$.
 2. Derive the DIF butterfly: Addition/subtraction occurs *first*, and twiddle factor multiplication is applied to the difference branch *afterwards*.
 3. Contrast DIT and DIF architectures:
 - DIT: Bit-reversed inputs $\rightarrow$ Natural frequency outputs.
 - DIF: Natural time inputs $\rightarrow$ Bit-reversed frequency outputs.
 4. Show that cascading a DIT-FFT and DIF-FFT (or vice-versa) allows back-to-back fast convolution without any bit-reversal sorting between stages.
 5. In-place memory mechanics for DIF butterflies.
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Decimation-in-Frequency (DIF) butterfly equations mathematically.
 2. **Draw** complete signal flow graphs for $N = 4$ and $N = 8$ DIF-FFTs.
 3. **Compare** DIT and DIF algorithms across computational complexity, butterfly structure, and memory ordering.
 4. **Compute** the Inverse DFT (IDFT) using standard forward FFT routines.
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2. MATHEMATICAL FOUNDATIONS

2.1 Derivation of the DIF Algorithm

Divide the N -point input sum into two halves of length $N/2$:

$$X[k] = \sum_{n=0}^{N/2-1} x[n]W_N^{nk} + \sum_{n=N/2}^{N-1} x[n]W_N^{nk}$$

Substitute $n \rightarrow n + N/2$ in the second summation:

$$X[k] = \sum_{n=0}^{N/2-1} x[n]W_N^{nk} + \sum_{n=0}^{N/2-1} x[n + N/2]W_N^{(n+N/2)k} = \sum_{n=0}^{N/2-1} [x[n] + x[n + N/2]W_N^{Nk/2}]W_N^{nk}$$

Since $W_N^{Nk/2} = (W_N^{N/2})^k = (-1)^k$:

$$X[k] = \sum_{n=0}^{N/2-1} [x[n] + (-1)^k x[n + N/2]]W_N^{nk}$$

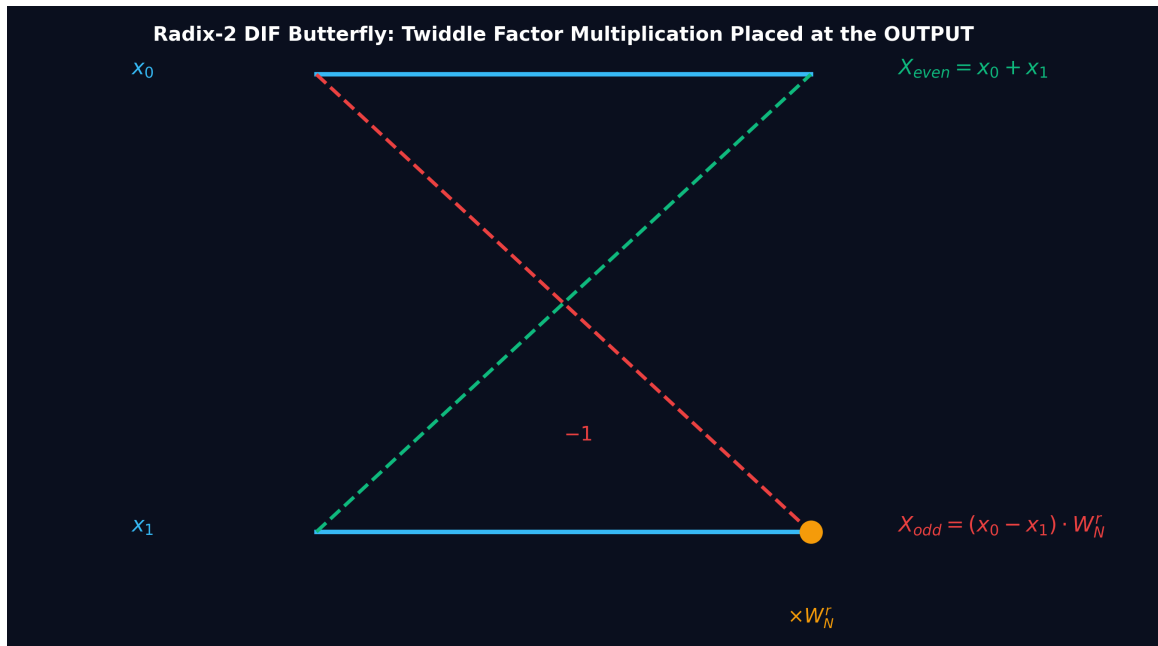
- For even frequency indices ($k = 2r$):

$$X[2r] = \sum_{n=0}^{N/2-1} [x[n] + x[n + N/2]] W_N^{2nr} = \sum_{n=0}^{N/2-1} [x[n] + x[n + N/2]] W_{N/2}^{nr} = \text{DFT}_{N/2} x[n] + x[n + N/2]$$

- For odd frequency indices ($k = 2r + 1$):

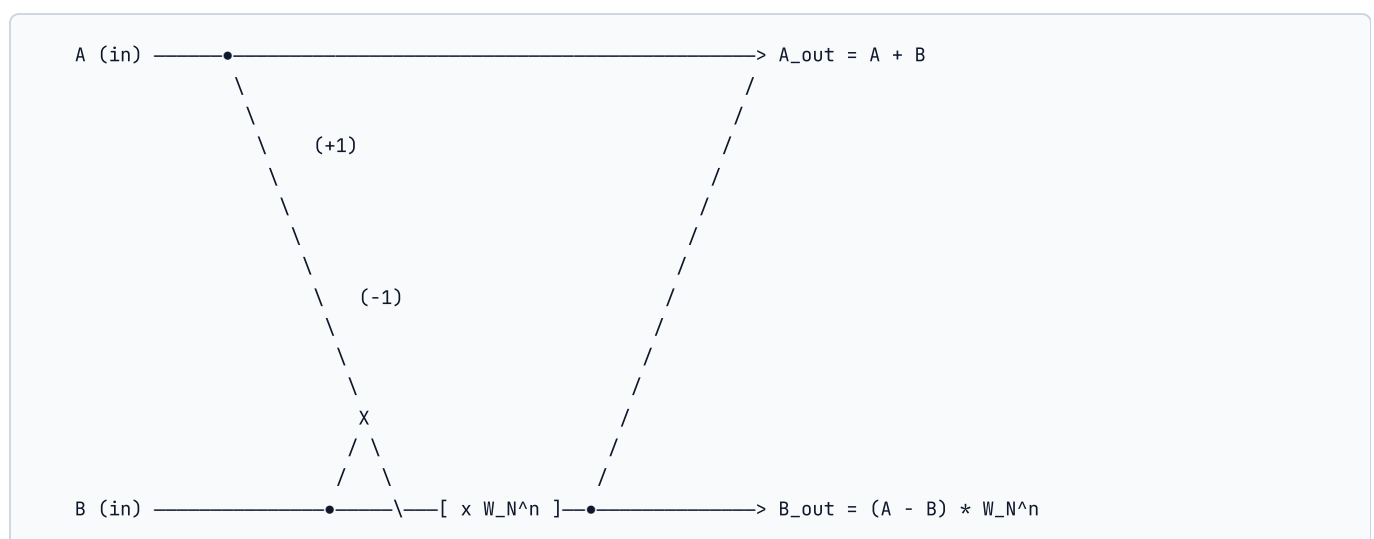
$$X[2r + 1] = \sum_{n=0}^{N/2-1} [x[n] - x[n + N/2]] W_N^n W_{N/2}^{nr} = \text{DFT}_{N/2} (x[n] - x[n + N/2]) W_N^n$$

2.2 The DIF Butterfly Processing Element



$$A_{\text{out}} = A + B$$

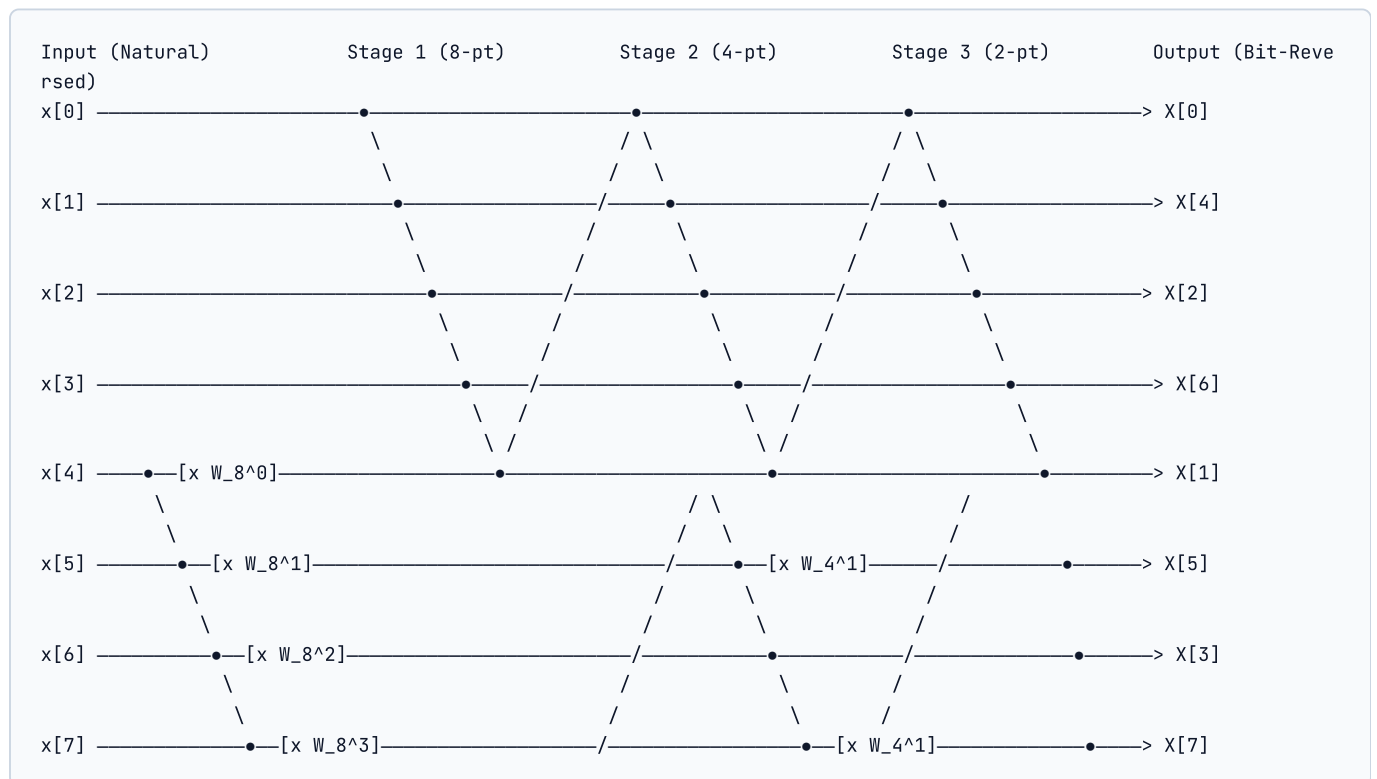
$$B_{\text{out}} = (A - B) W_N^n$$



2.3 Comprehensive Comparison: DIT vs. DIF FFT

Feature	Radix-2 DIT (Cooley-Tukey)	Radix-2 DIF (Sande-Tukey)
Decimation Domain	Time Domain (Even / Odd samples)	Frequency Domain (Even / Odd bins)
Input Ordering	Bit-Reversed Order	Natural Sequential Order
Output Ordering	Natural Sequential Order	Bit-Reversed Order
Butterfly Structure	$A \pm W_N^r B$ (Multiply <i>before</i> add/sub)	$(A + B)$ and $(A - B)W_N^r$ (Multiply <i>after</i> sub)
Complex Multiplications	$\frac{N}{2} \log_2 N$	$\frac{N}{2} \log_2 N$
Complex Additions	$N \log_2 N$	$N \log_2 N$

2.4 Complete 8-Point DIF-FFT Signal Flow Graph (SFG)



3. WORKED NUMERICAL EXAMPLES

Example 11.1: 8-Point DIF-FFT Step-by-Step

Problem: Compute the 8-point DFT of $x[n] = \underset{\uparrow}{1}, 1, 1, 1, 0, 0, 0, 0$ using Radix-2 DIF-FFT.

Solution:

- **Stage 1 (Twiddles W_8^n , ; $n = 0, 1, 2, 3$):**
 - $v_1[0] = x[0] + x[4] = 1 + 0 = 1$
 - $v_1[1] = x[1] + x[5] = 1 + 0 = 1$

- $v_1[2] = x[2] + x[6] = 1 + 0 = 1$
- $v_1[3] = x[3] + x[7] = 1 + 0 = 1$
- $v_1[4] = (x[0] - x[4])W_8^0 = (1 - 0)(1) = 1$
- $v_1[5] = (x[1] - x[5])W_8^1 = (1 - 0)(0.7071 - 0.7071j) = 0.7071 - 0.7071j$
- $v_1[6] = (x[2] - x[6])W_8^2 = (1 - 0)(-j) = -j$
- $v_1[7] = (x[3] - x[7])W_8^3 = (1 - 0)(-0.7071 - 0.7071j) = -0.7071 - 0.7071j$
- **Stage 2 (Twiddles $W_4^0 = 1, W_4^1 = -j$):**
 - Upper half:
 - $v_2[0] = v_1[0] + v_1[2] = 1 + 1 = 2$
 - $v_2[1] = v_1[1] + v_1[3] = 1 + 1 = 2$
 - $v_2[2] = (v_1[0] - v_1[2])W_4^0 = (1 - 1)(1) = 0$
 - $v_2[3] = (v_1[1] - v_1[3])W_4^1 = (1 - 1)(-j) = 0$
 - Lower half:
 - $v_2[4] = v_1[4] + v_1[6] = 1 - j$
 - $v_2[5] = v_1[5] + v_1[7] = (0.7071 - 0.7071j) + (-0.7071 - 0.7071j) = -1.4142j$
 - $v_2[6] = (v_1[4] - v_1[6])W_4^0 = (1 - (-j))(1) = 1 + j$
 - $v_2[7] = (v_1[5] - v_1[7])W_4^1 = (1.4142)(-j) = -1.4142j$
- **Stage 3 (Twiddle $W_2^0 = 1$):**
 - $X_{br}[0] = v_2[0] + v_2[1] = 4 \implies X[0] = 4$
 - $X_{br}[1] = v_2[0] - v_2[1] = 0 \implies X[4] = 0$
 - $X_{br}[2] = v_2[2] + v_2[3] = 0 \implies X[2] = 0$
 - $X_{br}[3] = v_2[2] - v_2[3] = 0 \implies X[6] = 0$
 - $X_{br}[4] = v_2[4] + v_2[5] = 1 - 2.4142j \implies X[1] = 1 - 2.4142j$
 - $X_{br}[5] = v_2[4] - v_2[5] = 1 + 0.4142j \implies X[5] = 1 + 0.4142j$
 - $X_{br}[6] = v_2[6] + v_2[7] = 1 - 0.4142j \implies X[3] = 1 - 0.4142j$
 - $X_{br}[7] = v_2[6] - v_2[7] = 1 + 2.4142j \implies X[7] = 1 + 2.4142j$

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) Derive the Radix-2 Decimation-in-Frequency FFT equations and construct the complete 8-point DIF signal flow graph. (10 Marks) **(b)** Compare Radix-2 DIT and Radix-2 DIF FFT algorithms in terms of input/output data ordering, butterfly equations, and hardware complexity. (5 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Splitting summation into first/second halves and separating into even/odd frequency bins (4 Marks)
 - Butterfly equations $A + B$ and $(A - B)W_N^n$ (2 Marks)
 - Neatly drawn 8-point DIF SFG showing natural inputs, 3 stages, and bit-reversed outputs (4 Marks)
- **Part (b):**
 - Comprehensive comparison table covering all items listed in Section 2.3 (5 Marks)

5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

x = np.array([1, 1, 1, 1, 0, 0, 0, 0])
X = np.fft.fft(x)
print("8-Point DIF-FFT Output:")
for k in range(8):
    print(f"X[{k}] = {X[k].real:.4f} + {X[k].imag:.4f}j")
```